SMN2-RescueDiff: A Failure-Aware, Auditable Pipeline for RNA–Ligand Prioritization

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***Abstract*—Computational RNA-ligand screening can produce persuasive rankings even when an upstream structural assumption is wrong. We present SMN2-RescueDiff, an auditable pipeline for candidate generation and *SMN2* exon-7 RNA prioritization whose principal methodological contribution is an explicit evidence contract between modules: each transition preserves configuration, provenance, missingness, failure records, geometry checks, and claim boundaries. The production run used an unconditional autoregressive SELFIES Transformer, not diffusion or target-conditioned generation. Of 50,000 requested samples, 49,660 valid strings were emitted, 171 collapsed during canonicalization, 49,489 unique canonical molecules remained, 40,477 passed chemical filters, and 1,000 were exported. A controlled 50,000-sample token 3-gram baseline produced 915 valid and 163 filter-pass molecules. Real AutoDock Vina target docking completed 2,500 jobs for 500 molecules against five fixed models from the 8R62 NMR ensemble; target-conformer robustness is the primary computational screen. A retrospective audit then falsified the original HuRDA structural selectivity analysis: one target-centered box had been applied to independently positioned decoy receptors, 81 of 97 receptors had under 10% of atoms inside the box, box coverage correlated with receptor mean score at Spearman $\rho = -0.847$, and receptor identity explained 98.8% of score variance. The 14,259 historical decoy jobs are therefore preserved but withdrawn from biological inference. Module 4 now supports receptor-specific boxes and fixed-box coverage gates. Redocking recovered a rank-1 pose below 2 Å for one of four RNA–ligand complexes, so calibration remains mixed. The result is not a validated drug or selectivity claim. It is a reusable computational-biology design for detecting, localizing, and containing invalid evidence before it becomes a biological conclusion.**

***Index Terms*—Auditability, evidence provenance, molecular generation, RNA docking, SMN2, reproducibility, virtual screening.**

I. Introduction

SPINAL muscular atrophy (SMA) is a neuromuscular disorder caused by insufficient functional survival motor neuron protein, most often following disruption of the *SMN1* gene. The closely related *SMN2* gene can partially compensate, but inefficient inclusion of exon 7 limits production of stable full-length SMN protein. Increasing *SMN2* exon-7 inclusion is therefore an established therapeutic strategy [1], [2]. Risdiplam demonstrates that RNA splicing can be modulated with a small molecule, while nusinersen and onasemnogene abeparvovec illustrate complementary modalities. These clinical advances do not make computational discovery simple: RNA is dynamic, charged, conformationally heterogeneous, and capable of presenting similar recognition surfaces across the transcriptome [3], [4], [5].

Generative models can propose molecules outside a fixed screening library, but validity and novelty do not imply target binding. This distinction is especially important for RNA, where shortcut learning, limited labeled data, receptor flexibility, and scoring functions developed for protein–ligand complexes can dominate apparent performance [6], [7], [8]. A computational workflow can also fail operationally while still producing complete tables: incompatible coordinate frames, silent score imputation, misaligned schemas, or selection on the same score later called a validation endpoint can each create a plausible but circular result.

The original motivation for SMN2-RescueDiff was to connect target definition, a human RNA decoy atlas, molecule generation, docking, and candidate ranking. The historical name is retained for continuity, but the generator is an unconditional autoregressive Transformer; it is not a diffusion model and is not conditioned on *SMN2*. The present study centers a more general TCBB contribution: a *failure-aware evidence architecture* for multistage computational biology. Its unit of innovation is not a new docking energy function. It is the contract that controls how evidence moves between heterogeneous modules and how a failed assumption is prevented from becoming a stronger downstream claim.

The study asks four engineering and scientific questions. First, can a full generation-to-screening workflow preserve enough run-level provenance to reproduce its counts and identify missing jobs? Second, does a neural molecular language model improve usable sample yield over a simple controlled generator baseline? Third, can ensemble target docking be summarized without allowing one favorable conformer to dominate? Fourth, when a structural counter-screen fails a coordinate audit, can the workflow expose the failure, preserve the records, and withdraw unsupported selectivity conclusions? No part of this study tests binding, exon-7 inclusion, toxicity, safety, or efficacy.

A. Contributions

The main contributions are:

- a modular, configuration-driven pipeline with stable schemas, run folders, copied configurations, failure tables, and machine-readable summaries;
- a sequence-scale HuRDA atlas kept explicitly separate from its small structural subset;
- a five-conformer target screen ranked by worst-case and mean Vina scores rather than the single best conformer;

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TABLE I
EVIDENCE-CONTRACT REQUIREMENTS

Contract element	Required behavior
Identity	Stable molecule, receptor, conformer, and run identifiers
Provenance	Input paths/hashes, source metadata, copied configuration
Execution	Engine, status, dry-run flag, seed, parameters, output path
Missingness	Failure reason retained; no silent score imputation
Geometry	Receptor/ligand format and coordinate-box coverage validated
Interpretation	Numeric output labeled as proxy, calibrated result, or audit-only

- a coordinate-aware docking gate that derives a box per decoy receptor or rejects a fixed box with inadequate atom coverage;
- a post hoc falsification analysis that quantifies receptor-frame, zero-score, panel-size, and molecular-size effects without altering the historical observations; and
- a release containing the previously missing 800-job cross-target campaign, benchmark-control files, generator checkpoint, controlled baseline, audit scripts, and response ledger.

II. MATERIALS AND METHODS

A. Pipeline architecture and evidence contracts

SMN2-RescueDiff is organized into six connected modules (Fig. 1). Each module writes versionable artifacts rather than passing transient in-memory state. A downstream module must be able to identify the exact input table, configuration, run name, status field, and whether a score is real or a dry-run placeholder. This is the pipeline's evidence contract (Table I). A failed contract does not receive a neutral or favorable numeric replacement; the corresponding record is marked failed or missing.

B. Module 1: SMN2 target definition

Module 1 defined exon-7-centered windows from a configured genome and annotation, converted DNA sequence to RNA, and emitted FASTA, feature, and metadata files. It calculated nucleotide frequencies, GC content, k-mer counts, simple splice-site/U1-complementarity features, and distances from exon boundaries. These quantities define a reproducible target profile; they are not evidence of small-molecule activity.

The structural target panel used the first five deposited models of the 8R62 solution-NMR ensemble. These are five conformations of one risdiplam-associated RNA complex, not five independent biological targets. This narrow ensemble probes local conformer sensitivity but does not represent all accessible *SMN2* RNA folds.

C. Module 2: HuRDA sequence atlas and structural subset

HuRDA-180K was designed as a sequence-level counter-screening resource. It contains over 180,000 candidate motifs from 5′ and 3′ splice sites, exon- and intron-adjacent windows, *SMN2*-like sequences, mutated/scrambled controls, and bias-matched transcript windows. Each record has a conservative category label, sequence features, similarity scores, and train/validation/test or holdout assignment.

The sequence atlas must not be confused with structural docking coverage. One hundred motifs were prepared as predicted RNA receptors; 97 formed the common historical candidate matrix. Predicted structures do not establish native folds, accessibility, expression, or binding. The atlas supplies candidate decoys for staged experiments, not a transcriptome-wide safety screen.

D. Module 3: molecular generation and triage

The production generator was a decoder-only Transformer trained autoregressively on SELFIES tokens with causal masking. Training used 149,534 valid ChEMBL/ZINC-derived rows from 150,000 configured records. The configured R-BIND and SMMRNA sources contributed zero rows to this historical run. Known splicing modifiers were held out as benchmark controls and were not used for model training. Exact upstream database release identifiers and redistribution licenses could not be recovered with sufficient certainty, an end-to-end reproducibility limitation.

Generated strings were converted to canonical SMILES with RDKit, deduplicated at the canonical graph level, and filtered for molecular weight, lipophilicity, hydrogen-bond counts, rotatable bonds, polar surface area, Lipinski criteria, PAINS alerts, novelty, and a documented synthetic-accessibility proxy. A fingerprint-based diversity selector created the docking export. These are cheminformatic prioritization features, not RNA-activity labels.

For a controlled baseline, the same configured training-source mixture and requested sample count were supplied to a token 3-gram SMILES generator. The comparison measures valid and filter-pass yield under the two representations/architectures. It does not isolate every design choice and is not an external biological benchmark.

E. Module 4: target and decoy docking

Ligands were prepared in SDF and PDB for inspection and converted with Open Babel to PDBQT. Only an existing ligand PDBQT and receptor PDBQT can enter AutoDock Vina. Missing PDBQT files produce an explicit failed job. Target docking used identical Vina settings across the five target conformers. All reported scientific scores have `is_dry_run_score=false`.

For molecule i and target conformers $c = 1, \ldots, 5$, robustness statistics were

$$\bar{E}_i^T = \frac{1}{5}\sum_c E_{ic}^T, \quad E_{i,\mathrm{worst}}^T = \max_c E_{ic}^T, \tag{1}$$

with lower Vina values considered more favorable within this protocol. The primary ranking requires all five jobs and orders candidates by $E_{i,\mathrm{worst}}^T$, then $\bar{E}_i^T$, then conformer range and standard deviation. This prevents one unusually favorable conformer from controlling advancement.

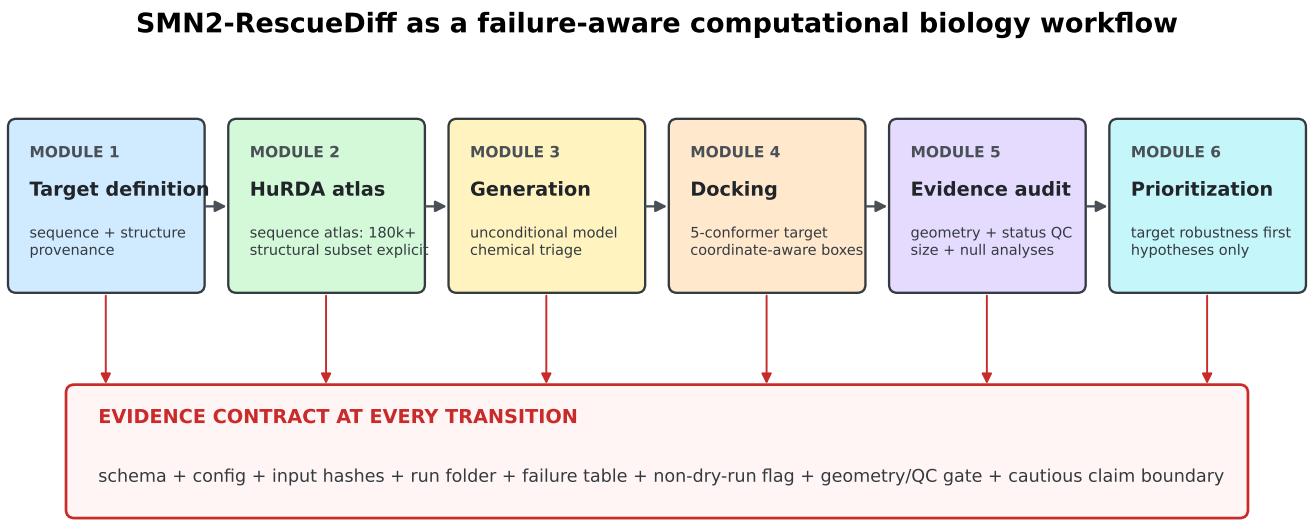


Fig. 1. SMN2-RescueDiff architecture. The contribution emphasized in this revision is the horizontal evidence contract: data schemas, configuration, provenance, failure records, score origin, geometry QC, and claim boundaries accompany every stage.

The original decoy implementation reused the target box for all receptors. The corrected implementation supports `per_receptor_bounds`: for receptor atom coordinates $\mathbf{x}_1, \ldots, \mathbf{x}_n$ and padding p, the box center and size are

$$\mathbf{c} = \frac{\min_j \mathbf{x}_j + \max_j \mathbf{x}_j}{2}, \qquad \mathbf{s} = \max_j \mathbf{x}_j - \min_j \mathbf{x}_j + 2p. \quad (2)$$

A fixed box can alternatively be used only when its receptor atom coverage exceeds a configured minimum. The current preset requires complete receptor-specific coverage and writes a per-receptor box-QC table before docking.

F. Historical Module 5 scores and primary analysis rule

The historical Stage C ranking was lexicographic and in practice was controlled by a continuous target–decoy gap. SRSI was a separate standardized weighted score that included target Vina score, the same target score inside gap and contact-derived terms, chemical quality, and penalties. They were not one rule. Because the decoy scores fail the coordinate audit and target evidence was counted in correlated terms, neither Stage C nor SRSI is the primary analysis here.

The SRSI is retained as a reproducibility artifact and labeled a heuristic. Its former "splicing-rescue proxy" is renamed the *geometric contact proxy*. It used global atom-pair counts and fixed fractional assignments rather than nucleotide-mapped functional contacts; several terms were saturated or nearly constant. It has not been calibrated against exon-7 inclusion and carries no mechanistic interpretation. The primary rule in this paper is the prespecified five-conformer target robustness ranking above.

G. Post hoc falsification analyses

For each historical decoy receptor, atom coordinates were compared with the original target-centered box. We calculated receptor atom coverage, exact-zero score fraction, receptor-level mean score, and Spearman correlation between coverage and score. One-way variance decomposition estimated the fraction of total score variance associated with receptor or ligand identity. Historical scores were not modified.

To inspect exact zeros, 250 zero-score and 250 nonzero-score jobs were sampled with seed 42. The minimum Euclidean distance between any ligand pose atom and receptor atom was calculated. Sensitivity analyses recomputed mean gaps after excluding zeros, conservative gaps after removing the strongest one or three decoy receptors, and worst-decoy statistics across 2,000 randomly sampled panels of 5, 10, 20, or 50 receptors. These analyses diagnose the historical experiment; they do not repair it.

Molecular-size sensitivity used RDKit heavy-atom count and molecular weight. A Vina ligand-efficiency descriptor was defined as $-\bar{E}_i^T / N_i^{\mathrm{heavy}}$. A linear score-versus-weight model provided a residual ranking. Neither is a binding free energy.

H. Calibration and additional campaigns

The docking protocol was evaluated on four deposited RNA–ligand complexes: 8R62/risdiplam, 8R63/branaplam, 8R8P/SMN-CX, and 5KX9/a ribocil analogue. Five seeds and retained modes distinguished sampling from top-rank scoring. A rank-1 pose below 2 Å was the prespecified recovery criterion.

A 40-molecule by 20-receptor non-RNA campaign was also released with all 800 scores, logs, poses, receptor inputs, and provenance. It tests pipeline execution and provides an exploratory promiscuity audit. Scores across heterogeneous proteins and RNA are not assumed to be commensurate binding energies. Risdiplam and branaplam target scores are released as held-out descriptive references. Their 2×100 decoy run shares the historical box defect and is audit-only.

TABLE II
PRODUCTION GENERATION FLOW

Stage	Count
Samples requested	50,000
Unique valid strings emitted	49,660
Canonicalization collapses	171
Unique canonical molecules	49,489
Unique chemical-filter failures	9,012
Retained unique molecules	40,477
Total rejection records	9,183
Docking export	1,000

III. RESULTS

A. Generation accounting and controlled baseline

The production sampler requested 50,000 samples and emitted 49,660 unique valid strings. Canonicalization collapsed 171 strings onto an already observed graph, leaving 49,489 unique canonical molecules. Chemical filters rejected 9,012 unique molecules and retained 40,477; the 9,183-row rejection table includes both those failures and the 171 canonical duplicates. Diversity selection exported 1,000 molecules (Table II; Fig. 2). This resolves the previously ambiguous arithmetic and gives each percentage a named denominator.

The 3-gram baseline produced 915 RDKit-valid records (1.83% of requested samples), 535 unique valid molecules, and 163 filter-pass molecules (0.326%). The production Transformer/SELFIES route produced valid and filter-pass fractions of 99.32% and 80.954%, respectively. Representation and architecture differ together, so this experiment shows a practical yield advantage for the production route, not that a specific neural component learned RNA recognition.

B. Five-conformer target screen

All 2,500 planned target jobs completed for 500 molecules and five 8R62 models. Across molecules, the median mean score was -7.193 kcal/mol and the median worst-conformer score was -7.018 kcal/mol. MolGen_038292 was ranked first by the robustness rule, with a mean of -8.929, worst score of -8.590, standard deviation 0.306, and range 0.776 kcal/mol (Fig. 3). These numbers identify a reproducible Vina ranking within one narrow receptor ensemble; they do not establish binding.

The previous matched-control analysis selected a top 40 using a rule containing target score and then tested target score. Its mean enrichment was 0.729 kcal/mol, but a target-only top 40 produced 1.016 kcal/mol and overlapped 35 of 40 molecules. Moreover, 400 candidate–control assignments reused 183 controls. The result is therefore reported only as within-cohort selection behavior. It is not independent evidence of prediction, SRSI utility, or selectivity.

C. Structural decoy audit falsifies the historical inference

The historical decoy table contains 14,259 completed, non-dry-run Vina rows for a complete 147×97 matrix. Completeness did not imply valid comparison. Eighty-one of 97 receptors had less than 10% of atoms inside the fixed target box. Coverage correlated strongly with receptor mean score ($\rho = -0.8469$, $p = 8.36 \times 10^{-28}$; Fig. 4). Receptors below 10% coverage averaged -1.193 kcal/mol, whereas the remainder averaged -6.886 kcal/mol. Receptor identity explained 98.83% of score variance and ligand identity 0.132% (Table III).

TABLE III
HISTORICAL DECOY FALSIFICATION AUDIT

Diagnostic	Result
Completed historical jobs	14,259
Receptors / ligands	97 / 147
Receptors with $< 10\%$ atoms in box	81
Exact-zero scores	4,815 (33.77%)
Coverage–mean-score Spearman ρ	-0.8469
Variance associated with receptor	98.83%
Variance associated with ligand	0.132%
Strongest receptor for all ligands	HuRDA_000041
All observed decoy weights	1.0

Exact zeros were structural warning signals, not missing values. In the seeded pose sample, zero-score poses had a median minimum ligand–receptor distance of 32.65 Å, compared with 2.58 Å for nonzero poses. The median historical mean gap decreased from 5.590 to 4.573 kcal/mol when exact zeros were excluded. HuRDA_000041 was the strongest-scoring decoy for all 147 ligands. Removing it changed the median conservative gap from -1.950 to -0.502 kcal/mol and made 12 of 147 gaps nonnegative; removing the three strongest receptors produced a median of -0.098 and 54 nonnegative gaps. Worst-decoy behavior also changed substantially with random panel size (Fig. 5).

All historical decoy weights were 1.0, so the former "biologically weighted" gap was an unweighted mean. The previously reported finding that all conservative gaps were negative cannot be interpreted as true poor selectivity: it is dominated by receptor placement. Conversely, the favorable mean gaps cannot be interpreted as selectivity either. Both directions of biological inference are withdrawn pending corrected receptor-specific redocking.

D. Corrected candidate descriptors and size sensitivity

RDKit recomputation corrected MolGen_038292 to molecular weight 377.448 Da, QED 0.8563, cLogP 2.7055, two donors, six acceptors, three rotatable bonds, tPSA 79.38 Å^2, and 28 heavy atoms. Historical copy stating 504 Da and QED 0.498 was wrong. The corrected historical stronger-decoy quantity is 1/97 = 0.010309 as a fraction, or 1.0309%; because the decoy campaign is invalid, even the corrected value is audit-only.

Mean target score correlated with molecular weight at $\rho = -0.5054$ ($p = 9.06 \times 10^{-34}$; Fig. 6). MolGen_038292 had Vina ligand efficiency 0.3189 and ranked 136 of 500 by that descriptor. Its molecular-weight-adjusted score residual was -1.344 kcal/mol, rank 3 of 500. Thus its target rank is not explained only by molecular weight, but raw Vina score cannot be called a stronger binding affinity or clinical advantage.

Risdiplam and branaplam were held out from training. A separately released real Vina target run produced scores of -8.591 and -8.321 kcal/mol, respectively, against one prepared target receptor. Those values are descriptive references under a different receptor design than the five-conformer candidate

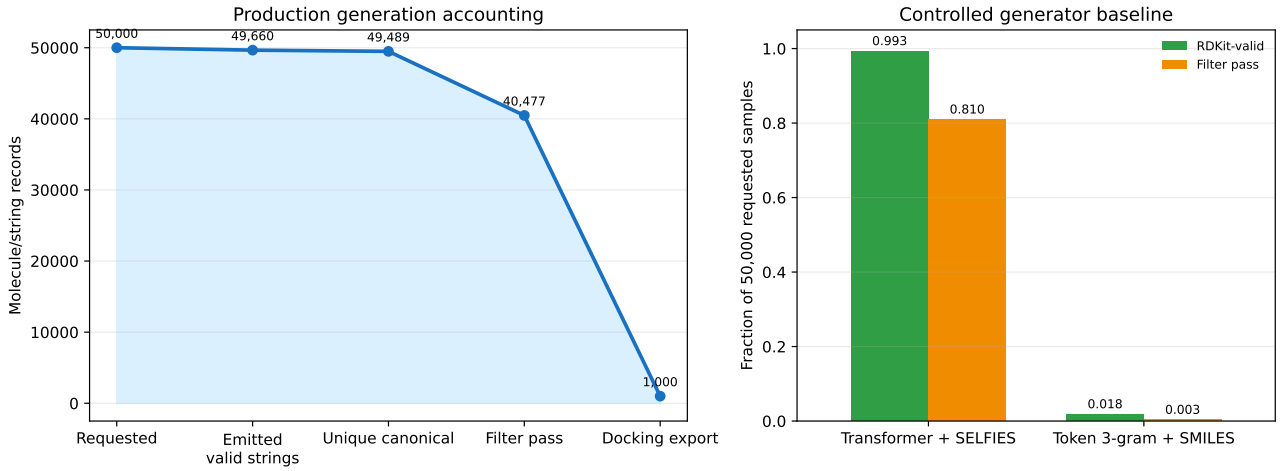


Fig. 2. Left: mutually reconciled production-generation flow. Right: controlled 50,000-sample engineering baseline. Transformer/SELFIES produced 49,660 valid and 40,477 filter-pass records; token 3-gram/SMILES produced 915 and 163. The comparison is not a biological validation.

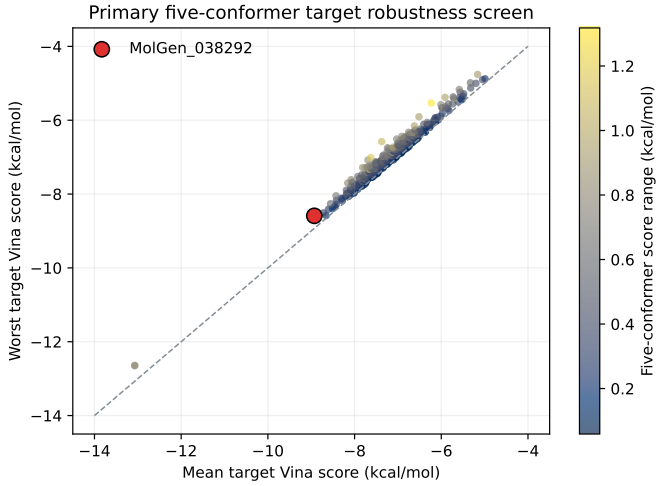


Fig. 3. Mean and worst-conformer target Vina scores for 500 molecules. Color is the five-conformer range. Advancement requires all five jobs and prioritizes the worst score before the mean.

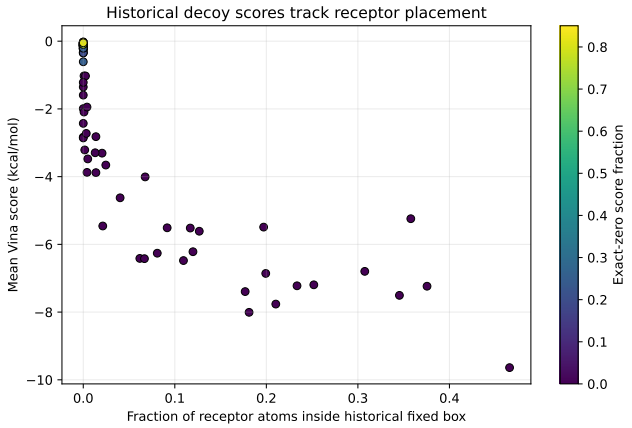


Fig. 4. Historical decoy scores track the fraction of each receptor inside the target-centered box. Color denotes exact-zero score frequency. This geometry dependence invalidates target–decoy selectivity interpretation of the campaign.

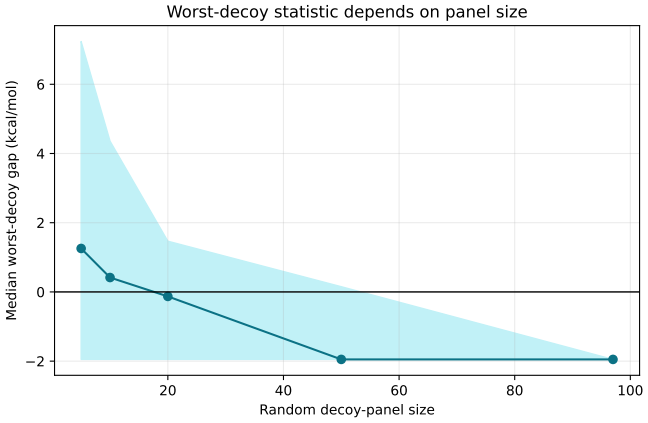


Fig. 5. Historical worst-decoy statistic as a function of randomly sampled panel size. The dependence is descriptive because the underlying fixed-box experiment is invalid for selectivity inference.

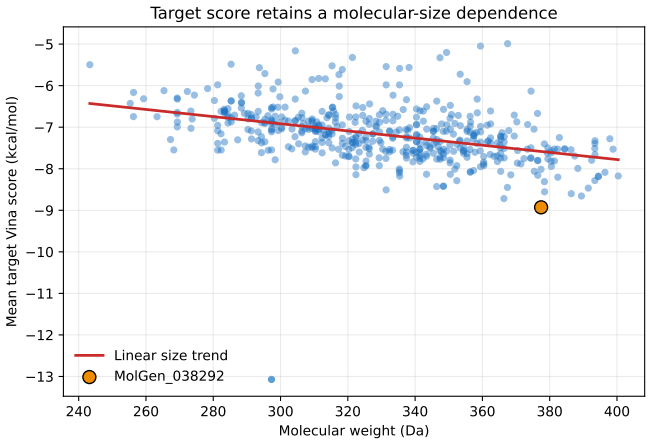


Fig. 6. Molecular weight retains a substantial association with mean target Vina score. The line is a post hoc linear trend used for residual sensitivity, not an affinity calibration.

summary. No statement that a generated molecule is clinically or biophysically superior is made.

TABLE IV
RNA–LIGAND REDOCKING CALIBRATION

Ligand	Rank-1 RMSD	Best mode RMSD	Rank-1 < 2 Å
Risdiplam	12.587	2.478	No
SMN-CX	7.169	1.856	No
Ribocil analogue	6.173	4.384	No
Branaplam	1.307	1.137	Yes

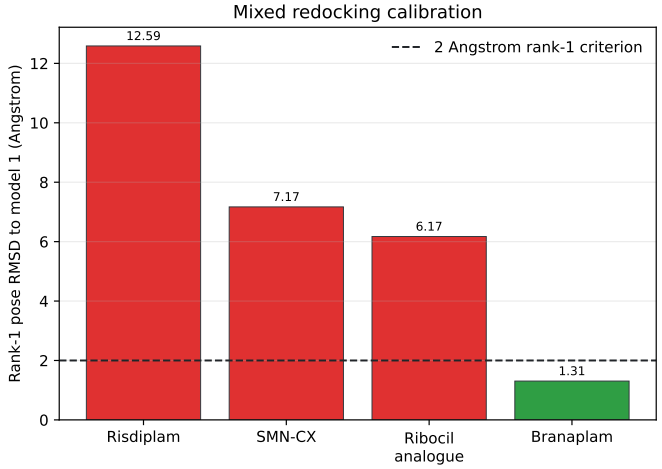


Fig. 7. Rank-1 pose RMSD for four deposited RNA–ligand complexes. The dashed line is the prespecified 2 Å criterion.

E. Calibration and released auxiliary campaigns

Redocking was mixed (Table IV; Fig. 7). Branaplam achieved a rank-1 pose at 1.307 Å from deposited model 1; risdiplam, SMN-CX, and the ribocil analogue had rank-1 RMSDs of 12.587, 7.169, and 6.173 Å. Lower-ranked modes reached 2.478 Å for risdiplam and 1.856 Å for SMN-CX, distinguishing sampling from ranking failure. One of four rank-1 successes is weak calibration and does not validate prospective screening.

The previously uncheckable cross-target claim is now supported by an 800-row table: 40 molecules, 20 receptors, 800 completed real Vina jobs, with all logs and poses released. This closes an artifact gap but does not validate cross-class score comparability. The complete risdiplam/ branaplam 2×100 campaign is also released; its decoy portion is quarantined because it shares the fixed-box defect.

IV. Discussion

A. The computational-biology innovation

The strongest outcome of this project is not that a molecule "won" a docking table. It is that the pipeline retained enough structure to show why an apparently complete selectivity result was invalid. The original matrix had real Vina jobs, valid PDBQT inputs, no imputation, and complete molecule–decoy coverage. Conventional completeness checks therefore passed. The missing check was geometric: receptors generated in different coordinate frames cannot inherit a target-centered search box. Once box coverage was measured, the source of the signal became quantifiable rather than speculative.

This suggests a broader TCBB design principle. In multi-stage workflows, schema validation alone is insufficient; evidence contracts must include domain invariants. For docking, those invariants include coordinate frames, atom coverage, receptor preparation, and score origin. For generation, they include canonicalization denominators and held-out benchmarks. For ranking, they include independence between the selection criterion and validation endpoint. Encoding such checks makes failure a first-class, reproducible output.

The new Module 4 behavior operationalizes that principle. Each decoy can receive a box derived from its own receptor bounds, and a fixed box must pass a configurable coverage threshold. The run writes receptor-specific box QC alongside scores. This does not guarantee biological accuracy, but it prevents the exact coordinate mismatch found here. The same principle applies beyond docking: downstream modules should reject an upstream artifact that cannot satisfy its scientific preconditions.

B. What succeeded, and what did not

The workflow succeeded as software and as an audit system. It trained and sampled a production generator, reconciled generation denominators, outperformed a simple baseline on usable yield, completed a five-model target campaign, ranked without a best-conformer shortcut, released a previously missing 800-job campaign, and surfaced the structural-decoy failure. These are meaningful computational-engineering results.

The study did not demonstrate predictive RNA selectivity or exon-7 rescue. Historical mean and conservative gaps are unusable for that purpose. Target-only matched-control behavior is partly selected by construction. SRSI combines correlated, uncalibrated terms and is a heuristic. Redocking is weak at rank 1. AutoDock Vina was parameterized primarily for protein–ligand systems, and its values are comparative scores rather than measured free energies [14], [15], [7], [8].

This candid boundary strengthens rather than erases the pipeline's contribution. A computational platform intended for iterative science must support negative audits and claim withdrawal, not only successful runs. Preserving the historical scores alongside their falsification analysis makes the error inspectable and prevents retroactive data cleaning from disguising it.

C. Ranking implications

Target-conformer robustness is the only primary ranking in this revision. It is intentionally modest: it prioritizes molecules whose Vina scores are consistently favorable across five models of one RNA ensemble. It does not use the invalid decoy panel, saturated contact terms, or target-score enrichment as independent validation.

The historical Stage C and SRSI outputs remain useful for software regression and ablation, but not for biological inference. Future ranking weights should be learned or selected using independent labels, and evaluation should compare the full rule with target-only, size-aware, and simple cheminformatic baselines on held-out compounds and scaffolds. Until then, "geometric contact proxy" is the maximum defensible label for the former rescue feature.

D. Limitations and required validation

This study has several nontrivial limitations.

First, no direct RNA-binding assay, cell-based exon-7 inclusion assay, toxicity measurement, pharmacokinetic result, or animal experiment was performed. The outputs are hypotheses for prospective testing.

Second, corrected receptor-specific HuRDA docking is configured but not completed in this report. No corrected selectivity number is substituted for the invalid historical one. A staged campaign should first test a small prespecified panel, inspect poses and box coverage, then expand to 500–1,000 hard decoys only if QC passes. Docking all 180,000 sequence motifs would conflate sequence coverage with unsupported structure prediction and waste computation.

Third, the target ensemble is the first five 8R62 NMR models associated with a ligand-bound complex. Alternative RNA folds, preparation choices, protonation/charge models, water/ion treatment, and box definitions may change ranks. The first five models are not independent targets.

Fourth, four redocking complexes are too few to estimate generalization. A prespecified benchmark of at least 20–30 structurally diverse RNA–ligand complexes, appropriate non-binders, multiple folds, and both pose and binder-ranking metrics is required. RNA-aware rescoring should be tested against Vina. These experiments are outstanding; no surrogate labels were fabricated.

Fifth, the generator baseline is deliberately simple. It combines a different representation and architecture and does not compare against matched GRU, random-library, or database-retrieval baselines. The production generator was not trained on RNA-focused rows and is not target conditioned. Its advantage here is usable sample yield, not demonstrated RNA-ligand learning.

Sixth, exact ChEMBL/ZINC releases, retrieval dates, and redistribution licenses for the historical subset could not be reconstructed. The checkpoint, tokenizer, configuration, source accounting, and hashes are released, but the third-party corpus is not redistributed. This prevents full regeneration from the release alone.

Finally, 190 rows in the merged target table have a blank legacy engine field despite completed non-dry-run provenance elsewhere. They were not silently relabeled. Future merges must enforce non-null engine metadata.

E. Prospective validation plan

A defensible next study has three locked stages. Stage 1 will run the receptor-specific-box HuRDA panel and reject any receptor that fails atom coverage, typing, or pose-contact QC. Stage 2 will benchmark Vina and at least one RNA-aware reranker on a predeclared external complex/binder set with scaffold-separated splits. Stage 3 will test a small, chemically diverse candidate set using direct binding and *SMN2* exon-7 inclusion assays, with risdiplam and branaplam as held-out reference controls. Only those data could support a rescue or therapeutic claim.

V. Conclusion

SMN2-RescueDiff demonstrates a reproducible, failure-aware architecture for RNA-ligand prioritization. The pipeline generated and triaged a large molecular cohort, completed a robust five-conformer target screen, and published run-level evidence across generation, docking, redocking, and auxiliary campaigns. More importantly, it exposed that a complete real- Vina decoy matrix was dominated by an incompatible search box, quantified the effect, preserved the failed evidence, and changed the software so that the same error is rejected prospectively.

The scientific claim is deliberately narrow. MolGen_038292 is a computationally prioritized target-docking hypothesis, not a drug. Historical HuRDA scores do not establish selectivity, SRSI does not establish splicing rescue, and mixed redocking does not validate the ranker. The contribution to computational biology is the auditable methodology that makes those boundaries visible and enforceable. In a field where multistage workflows can amplify one hidden assumption into a persuasive final rank, detecting and containing invalid evidence is itself a necessary form of computational rigor.

Acknowledgments

The author used ChatGPT, Claude, Codex, and Grammarly for code review, repository auditing, figure layout, and manuscript revision. These tools did not generate the historical docking calculations. Numerical claims in this revision were recomputed from released tables or logs, and no missing score was invented or imputed. The author reviewed the complete text and takes responsibility for its claims. The author thanks Dr. Joy Li (Duke University) for mentorship.

Code and Data Availability

The source and evidence release are available at https://github.com/sakshams-cyber/SMN2-RescueDiff. The audited evidence tree is immutable at commit `5e24251` under `reproducibility/tcbb_2026`. The `reproducibility/tcbb_2026` directory contains target and historical decoy inputs/scores, failure records, audit tables, redocking, the complete 40×20 campaign, benchmark-control v2 files, the production checkpoint/tokenizer/configuration, the 50,000-sample baseline, source manifests, and checksum files. The review-response ledger in the `docs` directory maps each evaluator request to an implemented correction, withdrawn claim, partial release, or outstanding experiment. The Overleaf package contains this source, all figure inputs, generated figures, a change report, and a reproducible build guide.

Declarations

Ethical approval: Not applicable. This computational study used no human participants, identifiable human data, or animal subjects.

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